MTH-202

Quiz-Final-Term

by Attiq Kundi

Lecture: 23 to onward

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1. In the proof by contradiction, we lead the assumption to___.

   Answer: absurdity

2. The predicate describing the initial state is called____.

   Answer: pre-condition of the algorithm

3. An integer n is prime number, if and only if for all positive integers r and s, if n= rs then____.

Answer: r = 1 or s = 1

4. If ‘f’ is a perfect square, then which of the following is the correct representation of ‘r’?

Answer: r = k^2 for some integer k

5. Which of the following is not a rational number?

Answer: √7

6. To prove by mathematical induction for
1+2+3+...+n= (n) (n+1)/2 for all positive integers, the basis step is____.

Answer: 1 = (1) (1+1)/2

7. GCD of (8, 12) will be____.

Answer: 4

8. The statement $4^n>3^n+4$ is true when____.

Answer: □ ≥ □

9. If n is any positive integer then $2^n \geq 2(n+1)$ is true for all_____.

Answer: $n \geq 3$

10. The sum of $(6-7\sqrt{2})$ and $(6+7\sqrt{2})$ is____.

Answer: 12

11. The sum of two integers is not an even integer when____.

Answer:

12. If n is an add integer then ‘3n+2’ is____.

Answer: odd

13. The sum of any rational number and any irrational number is irrational.

Answer: true

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14. A proof by___is based on the logical equivalence between a statement and its contrapositive.

Answer: contrapositive

15. For every prime number n, n+2 is prime.
Which of the following prime number disproves the above statement.

Answer: n=7

16. In basis property the first iteration of the loop is solved for?

Answer: I (0).

17. For all integers n>0
1+2+2^2+___+2^n=___.

Answer: 2^ (n-1)

18. In division algorithm ‘r’ stands for____.

Answer: remainder.

19. If n is any positive integer then the sequence 3+6+9+---- +3n =_____.

Answer: 3n(n+1)
2

20. For all positive real numbers a and b, if a<b, then____.

Answer: a^2 < b^2

21. The last step of describing algorithms formally is?

Answer: An end statement

22. If ‘n’ is an odd integer then n^3+n is____

Answer: even

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23. Proof by uses the equivalence p→q = -q→ □.

Answer: Contraposition

24. The product of an even and odd integer is____.

Answer: even

25. In the Direct Proof, we show that if the statement p is true, then the statement q is___.

Answer: True

26. (-1)^n = 1 for n =____.
(^ stands for power)

Answer: 4

27. Which of the following is true for n = 1?

Answer: 2^ (2n) – 1 by 3

28. Proof by____uses the equivalence p→q =(p∧ −q)→ □.

Answer: Contradiction

29. Proof by contradiction is a/an____method of proof.

Answer: Indirect

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30. An integer ‘n’ is____if it can be represented as a multiple of 2.

Answer: even.

31. The contrapositive of ‘5n+2 is odd, then n is odd’ is___.

Answer; “if n is even then 5n+2 is even”

32. Which of the following is not a method to prove the mathematical statements?

Answer: Permutation.

33. The sum of two integers is not an even integer when____.

Answer: one is even and one is odd.

34. The sum of two odd integer’s is____.

Answer: even

35. The predicate which describes the initial state is called____of the algorithm.

Answer: Pre-condition

36. Algorithm is a more general term in that term____refers to a particular programming language.

Answer: Program

37. In division algorithm the pre-condition is?

Answer: A is a nonnegative integer and d is a positive integer.

38. Basis and inductive steps are part of proof by____.

Answer: Mathematical induction

39. If A and B are disjoint sets then P (A B) =……

Answer: P (A)

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40. If B is a subset of A then P(A/B)= 1

Answer: True

41. $P(A^C/B)=1 - P(A/B)$

Answer: True

42. If A and B be events with P (A) =1/3. P (B) =1/4 and P (A intersection b)=1/6, then P(A U B) =_____.

Answer: 5/12

43. P (A/B) = P (A n b)/P (B), where_____.

Answer: P (B) > 0

44. Total degree of a graph G is_____.

Answer: 2(Number of edges of G)

45. Let $E(X^2)$ =6, u^2=2, then standard deviation is_____.

Answer: 2

46. If A and B be events with P (A) =1/3. P(B)=1/4 and P(A intersection b)=1/6, then P(A/B) = _____

Answer: 2/3

47. A closed walk is the walk that starts and ends at the same vertex

Answer: True

48. Total degree of G is an odd number

Answer: False

49. If X and Y are random variable, then E(aX) is equal to

Answer: aE (X)

50. A simple graph is a graph that does not have any loop.

Answer: True

51. Conditional probability of A given B is defined as_____.

Answer: P (A n B) / P (B)

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52. If X and Y are independent random variables and a and b are constant, then Var(aX+bY) is equal to

Answer: A^2 Var(X) + b^2Var(Y)

53. A random variable is also called a

Answer: Chance variable

54. If the random variable X denotes the number of heads when three distinct coins are tossed, then X assumes the value

Answer: 0,1,2,3

55. If X and Y are random variables, then E (X-Y) is equal to

Answer: E(X) - E(Y)

56. A random variable is also called a Stochastic variable

Answer: True

57. Which of the followings is the probability of getting a 6 when a dice is tossed?

Answer: 1/6

58. A random variable is also called a

Answer: Chance variable or stochastic variable

59. If E(5)=5, then find arithmetic mean will be

Answer: 5

60. Conditional probability of B given A is defined as_____.

Answer: P (A n B) / P (A)

61. If A is a subset of B then P(A I B) = P(A)/P(B)

Answer: True

62. A graph is regular if every vertex has the_____degree

Answer: Same

63. If the standard deviation of the data set is 4 then the value of the variance is

Answer: 16

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64. What is the probability of the number of one head when two fair coins are tossed?

Answer: 2/4

65. A simple graph is a graph that does not have any loop

Answer: True

66. If A and B are disjoint sets then P (A/B) =_____?

Answer: P (A)

67. A complete graph on n vertices is a simple graph in which every is connected to every other vertex.

Answer: True

68. There are three bus lines between A and B, and two bus lines between B and C. Find the number of ways a person can travel round trip by bus from A to C by way of B?

Answer: 6

69. Which of the followings is the product set A*B*C? where A= {a}, B={b}, and C={c,d}

Answer: {(a, b, c), (a, b, d)}

70. In how many ways can 6 people be seated on 6 available seats?

Answer: 720

71. Among 20 people, 15 either swim or jog or both. If 5 swim and 6 swim and jog, how many jog?

Answer: 16

72. Compute C(8 , 3)

Answer: 56

73. Let X= {1, 2, 3}, then 2-combinations of the 3 elements of the set X are_____?

Answer: {1, 2}. {1, 3} and {2, 3}

74. P (A n B) = P (A). P(A/B)

Answer: False

75. If B is a subset of A then P(A/B) = 1

Answer: True

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76. An expected value of a random variable is equal to its mean

Answer: True

77. If X and Y are independent random variables, then E(XY) is equal to

Answer: E(x) E(y)

78. A random variable is also called stochastic variable

Answer: True

79. If A and B are two independent events then P(A n b) =

Answer: P (A) P (B)

80. If X is a discrete random variable and f(x) is the probability of X, then the expected value of this random variable is equal to;

Answer: ∑xf(x)

81. The expected value E(X) is obtained by multiplying each value of X by its probability and taking the sum.

Answer: True

82. P (A, A) = .

Answer: 1

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83. Which of the followings is the factorial form of 5 . 4?

Answer: 5!/3!

84. The set of prime numbers is .

Answer: Infinite set

85. Let R be a relation on a set A. If R is symmetric then its compliment is_____.

Answer: Irreflexive

86. There are 4 bus lines between X and Y, and 5 bus lines between Y and Z, the number of ways a person can travel from X to Z?

Answer: 20

87. To prove by mathematical induction inductive step is_____.

Answer: If P (K) is true then P(K + 1) is true for all integers K > 1

88. Algorithm is a more general term in that the term _____refers to a particular programming language.

Answer: Program

89. The sum of rational and irrational number_____.

Answer: Is always an irrational number

90. Basis and Inductive steps are part of proof by .

Answer: Mathematical induction

91. Reduction and absurdum is another name of____.

Answer: Proof by contradiction

92. If one event can occur in n_1 ways a second event can occur in n_2 ways, a third event can occur in n_2 ways, how ways all of the events can occur in the order

Answer: N1+n2+n3

93. A non-zero integer d divides an integer n if and only if there exists an integer k such that_____

Answer: N=dk

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94. The sequence 1+2+3+4 --------- +n is equal to_____.

Answer: N (n+1)/2

95. The sum of two integers is not an even integer when_____.

Answer: One is even and one is odd

96. To prove by mathematical induction for 1+2+3+…+n= (n) (n+1)/2 for all positive integers, the basis step is_____.

Answer: 1=(1)(1+1)/2

97. To prove by mathematical induction for 1+5+9+…. + (4n-3) =n (2n-1) for all positive integers, the inductive step is true for n=k is_____.

Answer: 1+5+9+ + (4k-3) = k(2k-1)

98. Proof by_____uses the equivalence p -> q = ~q -> ~p

Answer: Contraposition

99. The predicate which describes the initial state is called the_____of the algorithm

Answer: Pre-condition

100. The last step of describing algorithms formally is?

Answer: An end statement

101. Which of the following is not a method to prove the mathematical statements?

Answer: Permutation

102. GCD of (330.156) is?

Answer: 6

103. Which of the following is not an irrational number?

Answer: Square root (4)

104. The sum of a rational and irrational number_____.

Answer: Is always an irrational number

105. If n is an odd integer then n^3+n is_____.

Even

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106. The____of a predicate variable is the set of all values that may be submitted in place of the variable

Answer: Domain

107. For all positive real numbers a and b if a < b, then_____.

Answer: A^2 < b^2

108. For every prime number n, n+2 is prime. Which of the following prime number disproves the above statement.

Answer: N=7

109. The set of prime number is_____.

Answer: Infinite set

110. An integer n is _ if it can be represented as a multiple of 2

Answer: Even

111. In inductive property the first iteration of the loop is solved for?

Answer: I (K)

112. Which of the following is not a method to prove the mathematical statements?

Answer: Permutation

113. In how many ways a student can choose a course from 2 science courses 3 literature courses and 5 art courses

Answer: 10

114. In the Direct Proof, we show that if the statement p is true, then the statement q is_____.

Answer: True

115. Which of the following is true for n = 2?

Answer: X^n – y^n is divisible by (x – y) ^ 2

116. The sum of any rational number and any irrational number is irrational.

Answer: True

117. The set of prime numbers is_____.

Answer: Infinite set

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118. The sum of two odd integers is_____

Answer: Even

119. How many multiples of 5 are there from 10 to 75?

Answer: 14

120. A bit string is a sequence of 0's and 1's.

Answer: True

121. The method of loop invariants is used to prove_____of a loop with respect to certain pre and post-conditions.

Answer: Correctness

122. _____is used to determine number of ordered or unordered arrangement of objects

Answer: Counting

123. While proofing by contradiction the equivalence _____is used.

Answer: P →q ≡ (p∧~q) →c

124. While proofing by contraposition the equivalence_____is used.

Answer: P →q≡ ~q→~p

125. Proof by contradiction is a/an____method of proof.

Answer: Indirect

126. Proof by contraposition is a/an____method of proof.

Answer: Indirect

127. If n is divisible by 5, then n^2 is divisible by 25 is the contraposition of_____.

Answer: N^2 is not divisible by 25, then n is not divisible by 5

128. The quantity a – d. q equals the?

Answer: Remainder

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129. In inductive step of proof by mathematical induction, the assumption is made for_____.

Answer: N=k+1

130. How many 4-bit string contain at least one '1'

Answer: 16

131. For all integers n>0

1+2+2^2+…..+2^n=_____

Answer: 2^(n+1) – 1

132. The chairs of an auditorium are to be labeled with two characters, a letter followed by a digit L letters are a and b and numbers are 3 and 5 what is the largest number of chairs that can be labeled differently?

Answer: 15

133. To prove by mathematical induction for 1+5+9+….+ (4n-3)=n(2n-1) for all positive integers, the basis step is_____.

Answer: 1=1(2(1)-1)

134. The predicate describing the final state is called_____.

Answer: Post condition of the algorithm

135. The predicate describing the initial state is called_____.

Answer: Pre-condition of the algorithm

136. If n is any positive integer then $2^n \geq 2(n+1)$ is true for all_____.

Answer: N > 3

137. The sum of (6 – 7 under root (2)) and (6 + 7 under root(2)) is

Answer: 12

138. GCD of (8, 12) will be_____.

Answer: 4

139. In division algorithm 'r' stands for _____

Answer: Remainder

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140. If the sum of two integers is even the difference would be _.

Answer: Even

141. Which of the following is true for n=1?

Answer: 2^ (2n) – 1 by 3

142. To prove by mathematical induction for 1+3+5+….+(2n-1)=n^2 for all positive integers, the basis step is_____.

Answer: LHS=p(1)=2x1=1, RHS = 1^2=1

143. The loop is correct if the four properties are true, one of the property is?

Answer: Eventual feisty of guard

144. The statement $n^2 > n + 3$ is true when_____.

Answer: N > 3

145. If 'n' is an odd integer then n^3+n is_____.

Answer: Even

146. (-1)^n=1 for n = (^standard for power)

Answer: 4

147. A boy can choose from 5 train services and 3 bus service to go to his hometown, How many total options he can choose from?

Answer: 8

148. The method of loop invariants is based on the people of_____.

Answer: Division algorithm

149. The statement $3^n < n!$ is true when_____.

Answer: n>6

150. In a city, the bus route numbers consists of a natural number less than 20, followed by one of the letters A, B, C, D, E and F. How many different bus routes are possible

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Answer: 114

151. The word algorithm refers to a step-by-step method for performing some action.

Answer: True

152. If n is an odd integer then 3n+2 is_____.

Answer: Odd

153. To prove by mathematical induction basis step is_____.

Answer: The proposition P (1) is true

154. There are 5 Chinese books and 6 English books, a student wants to select one optional book for both subject, total number of choices will be

Answer: 30

155. In division algorithm the pre-condition is?

Answer: A is a nonnegative integer and d is a positive integer

156. If there are 5 different optional courses in English and 3 different optional in Maths. Choices for a student who to take one optional.

Answer: 8

157. Number of combination of 3-bit string that contain exactly there '0' is_____.

Answer: 8

158. The sequence $1^3+2^3+3^3+\ldots\ldots +n^3$ =_____.

Answer: N (n+1) (2n+1)/6

159. If 'r' is a perfect square, then which of the following is the correct representation of 'r'?

Answer: R=k^2 for some integer k

160. Which of the following is the representation of an odd integer 'n'?

Answer: N=2k+1 for some integer

161. If n is any positive integer then the sequence 3+6+9+.....+3n=_____.

Answer: 3n (n+1)/2

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162. A predicate becomes_____when its variables are given specific values.

Answer: Sentence

163. The greatest common divisor (gcd) of two integers a and b is the largest integer that divides both a and b. it is called_____.

Answer: EUCLIDEAN ALGORITHM

164. A predicate is a sentence that contains a finite number of_____.

Answer: Variables

165. A proof by_____is based on the fact that either a statement is true or it is false but not both.

Answer: Contradiction

166. The_____of a predicate variable is the set of all values that may be substituted in place of the variable

Answer: Domain

167. Which of the following statement is true according to the division algorithm?

Answer: 17=15 x 3 + 2

168. The division by zero is allowed in mathematics

Answer: True

169. A student can choose a computer project from one of the two lists. The two lists contain 12 and 18 possible projects, respectively. How many possible projects are there to choose from?

Answer: 30

170. For integers a,b,c if a divides b and a divides c, then a divides (a + b)

Answer: True

171. Suppose there are different tea flavors and 5 different biscuit brands. A guest wants to take one tea and one brand of biscuit. How many choices are there for this guest?

Answer: 13

172. The sequence 1+2+3+4……+ n is equal to_____.

Answer: N (n+1)/2

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173. The product of two odd integers is_____.

Answer: Odd

174. A student has 3 optional courses in mathematics and 5 optional courses in physics, how many total choices for him to take one course?

Answer: 8

175. If the square of an integer is even, then that integer is_____.

Answer: Even

176. The first step of describing algorithm formally is?

Answer: The name of the algorithm, together with a list of input and output variables

177. Set of prime numbers is finite.

Answer: false

178. The sum of two irrational numbers in general_____.

Answer: Is always a rational number

179. Which of the following is the representation of an odd integer 'n'?

Answer: N=2k+1 for some integer 'k'

180. In how many ways can 6 people be seated on 6 available seats?

Answer: 12

181. The product of an even and odd integer is_____.

Answer: Even

182. In the direct proof, we show that if the statement p is true then the statement q is _____.

Answer: True

183. _____is the mathematics of counting and arranging objects

Answer: Combinatorics

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184. The Contrapositive of '5n+2' is odd, then n is odd is

Answer: If n is even then 5n+2 is even

185. For All positive real numbers a and b, if a < b, then_____.

Answer: A^2 < b^2

186. The rational number '2' is the product of which of the two irrational numbers.

Answer: Under root (2), under root (2)

187. The sum of two integers is even when_____.

Answer: Both are even or both are odd

188. Proof by contraposition is based on the logical equivalence between a statement and its.

Answer: Contrapositive

189. The sum of two irrational numbers must be an irrational number.

Answer: False

190. In how many ways a student can choose a course from 2 science courses, 3 literature courses and 5 art courses.

Answer: 10

191. The sequence 1+ 2+ 3+ 4 ----+ n is equal to_____.

Answer: n2/2

192. To prove by Mathematical induction basis step is_____.

Answer: The proposition P (1) is true

193. The____of a predicate variable is the set of all values that may be substituted in place of the variable.

Answer: Domain

194. The rational number '2' is the product of which of the two irrational numbers.

Answer: $\sqrt{2}$, $\sqrt{2}$

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195. Also let a predicate I(n), called the

loop invariant

196. **Basis Property:** The pre-condition for the loop implies that I(0) is true before the first iteration of the loop.

197. **Inductive property:** If the guard G and the loop invariant I(k) are both true for an integer $k \geq 0$ before an iteration of the loop, then I(k + 1) is true after iteration of the loop.

198. **Eventual Falsity of Guard:** After a finite number of iterations of the loop, the guard becomes false.

199. **Correctness of the Post-Condition:** If N is the least number of iterations after which G is false and I(N) is true, then the values of the algorithm variables will be as specified in the post-condition of the loop.

200. If the guard G and the loop invariant I(k) are both true for an integer $k \geq 0$ before an iteration of the loop, then I(k + 1) is?

Answer: True

201. The sum of two irrational numbers must be an irrational number.

Answer: False

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